information visualization *(continued)*
standards for 38–40
transforming data
categorizing data 54
changing datatypes 52–53
nesting 55–56
scaling 53–54
visualization from loaded data 65–67
inline functions 59–61
innerRadius() function 145
insert() function 102
interactive applications
brushing
brush events 281
creating brush 274–277
making brush user friendly 278–281
redrawing components 281
data dashboard example
bar chart 267
circle pack 267–268
loading data 265–266
overview 263–265
resizing based on screen size 268–269
spreadsheet 266–267
hover events 270–274
overview 261–263
interactive globes 7
interactive style
DOM manipulation 86–88
events 82–84
graphical transitions 84–86
using color
color mixing 90–92
discrete colors 92
for numerical data 92–94
overview 88–90
Internet Explorer 4
interpolators, line 129

J

JavaScript
array functions 33–34
method chaining 30–33
overview 29–30
JSON (JavaScript Object Notation) 37
json() function 38, 48

L

labels, component 298
layout() function 167, 169
layouts
components
adding labels 298
linking to scales 295–298
loading sample data 292–294
creating 284–291
histograms 140–142
overview 109–110
pack layouts 148–151
pie charts
creating ring chart 145
drawing pie layout 144–145
overview 142
transitioning 146–148
plugins for
Sankey diagram 163–169
word clouds 169
stack layout 158–162
trees 152–158
libraries, external 80–82
lighter() function 89
line charts
drawing from points 126–128
drawing many with multiple generators 128
interpolators 129
overview 124–126
line() function 129, 134
<line> element 21–23
linear() function 53, 62
linkDistance property 194
links() function 186
linkStrength property 194–195
loading data
formats and 48–50
overview 47–48
log() function 54

M

Mapbox 233
maps
canvas drawing 237
cartograms 238–239
drawing points on 212–213
finding data 206–207
GeoJSON 207–209
globes 221–226
graticule 217
hexbins 238
interactivity 215–216
overview 206
projection 209–210, 213–215
raster reprojection 238
satellite projection 226–227
scale 210–211
tile mapping 233–235
TopoJSON
file format 227
merging 229–231
neighbors 232–233
overview 227
rendering 228
transform zoom 237
Voronoi diagrams 238
zoom 217–221
measuring data 56–57
method chaining 30–33
minified versions 13
Miso Project 301
mixed-mode rendering techniques 310–316
moreLayouts() function 168
multilinestring 217
musical scores 8

N

nest() function 55, 140
nested data 35, 55–56
network data 36, 316–320
network visualization 7
force-directed layout
adding nodes 197–200
charge property 193
creating diagram 186–188
gravity property 193
linkDistance property 194
linkStrength property 194–195
manually positioning nodes 201–202
network measures 190–193
optimization 202
overview 185–186
removing nodes 197–200
SVG markers 188–190
updating network 195–196
overview 175–176
static diagrams
adjacency matrix 179–182
arc diagram 182–185
network data 177–178
node() function 86